Ex Undis, Utilitas

Utility from a Wave of Quantum Information

Timothy C. Burt

2025-12-01

1. Ad Astra
   - An Application by Any Other Name
2. Quantum Information Science and Technology
   - Waves Front
3. Quantum Very Long Baseline Interferometry
   - The Starlight is Going Through a Phase
4. Phase Closure
5. Support Material
   - References

A Night Sky in Grand Teton National Park



Orion at pre-dawn (Tim Burt, Nikon D7000, AF-S Nikkor 20mm, F1.8, 8s)

When I Heard the Learn'd Astronomer

When I heard the learn'd astronomer,
When the proofs, the figures, were ranged in columns before me,
When I was shown the charts and diagrams, to add, divide, and measure them,
When I sitting heard the astronomer where he lectured with much applause in the lecture-room,

How soon unaccountable I became tired and sick,
Till rising and gliding out I wander'd off by myself,
In the mystical moist night-air, and from time to time,
Look'd up in perfect silence at the stars.

— Walt Whitman, 1865

When I Heard the Learn'd Astronomer

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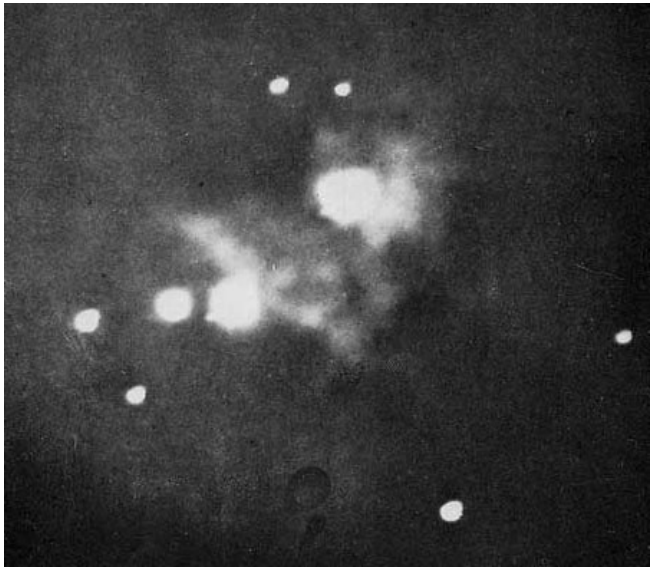
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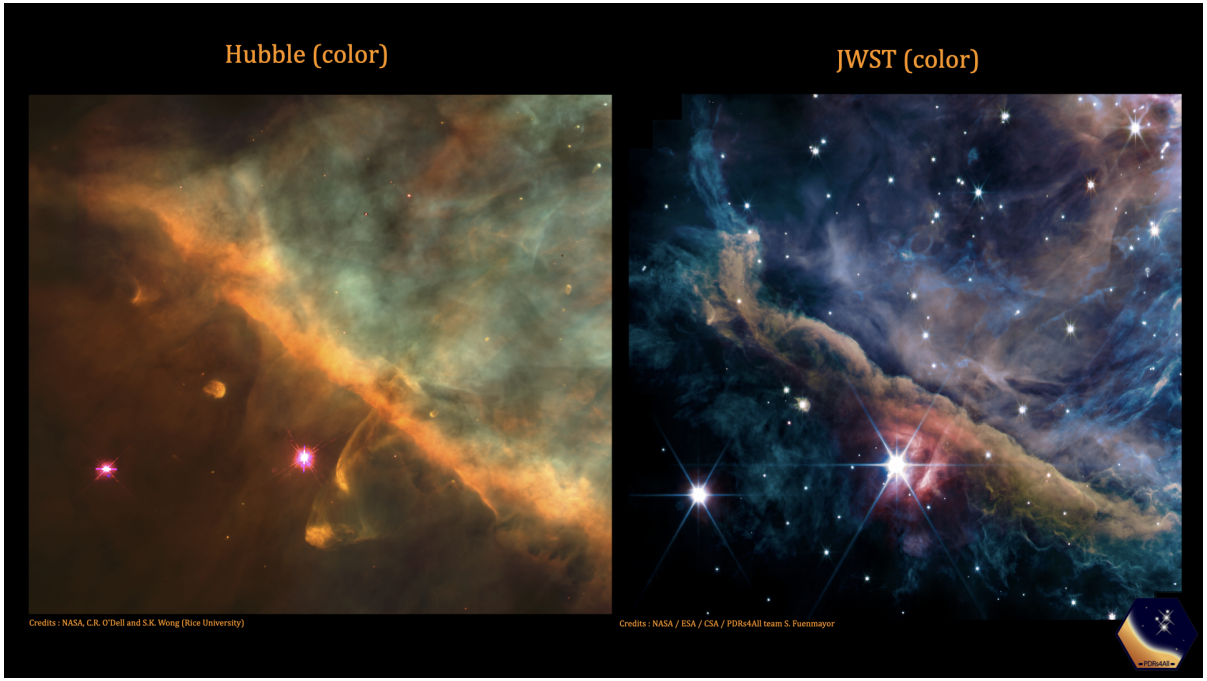
— Walt Whitman, 1865

1. Readers differ as to the extent the poem knocks science or extols mysticism.
2. In his book *Best Remembered Poems*, Martin Gardner says that this poem "is a familiar example" about "how science dulls our sense of nature's beauty".
3. In an analysis essay from 1988, Tim Burt (TB) wrote that the poem stands as a sad testament to the ability of the **classroom** to quell a student's sense of wonder.
4. The feeling that the reader gets is the same feeling the student gets; there is nothing new or different to do in astronomy.—TB 1988
5. The solution came when the student [. . .] redisovered his sense of wonder when he "wander'd off" in search of the love of the night sky and [. . .] found it exactly how and where it began, actively wondering under the stars.—TB 1988

Orion Through the Years and Better Eyes



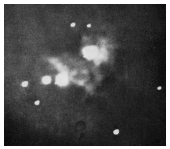
First image of Orion, taken by Henry and Anna Draper, 1880 [5]



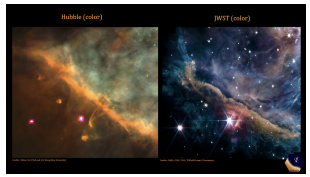
Orion from Hubble Space Telescope and James Webb Space Telescope [6]

Electromagnetic waves from roughly 1,500 light years away of the Orion Nebula (M24) are captured on film and digital detectors.

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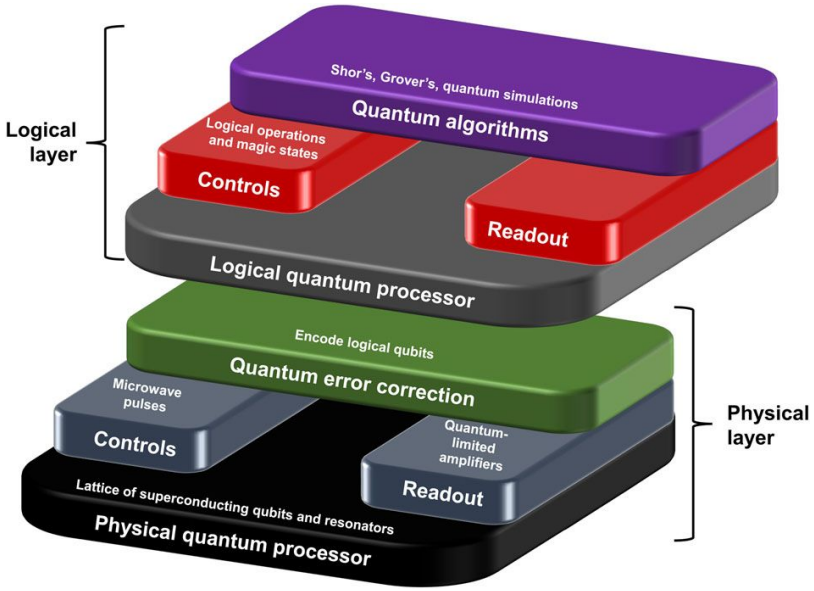
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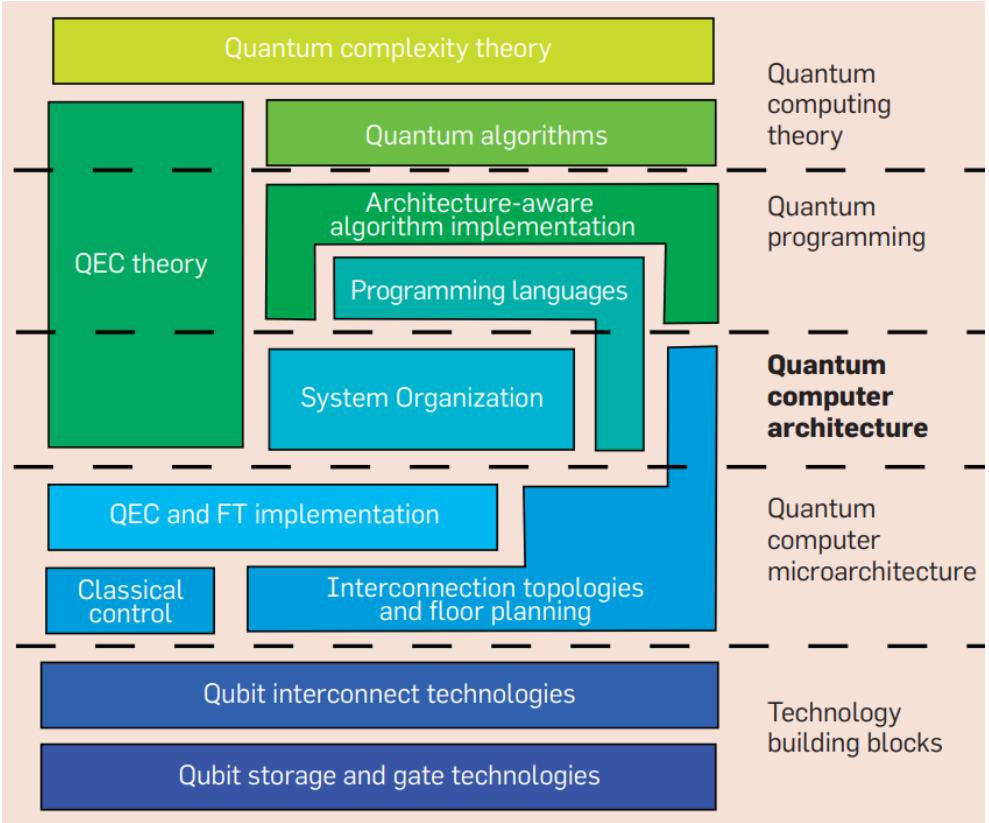
1. The Draper image used an 11-inch refractor telescope with an exposure time of 50 minutes. Two years later Draper captured a better image using a 137-minute exposure time. *First Photograph of the Orion Nebula*, in Cosmic Reflections (202)
2. Note the spikes around the brightest objects in the JWST image. They are evidence of the electromagnetic waves diffracting from linear features in the collection system itself.

The never-ending question: What is it good for?

Quantum Computing Stacks



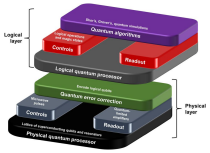
A notional stack for quantum computers from IBM [15]



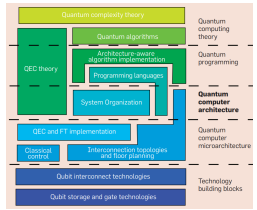
A more detailed view of quantum computer architecture from the Association for Computing Machinery [14]

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Quantum Computing Stacks



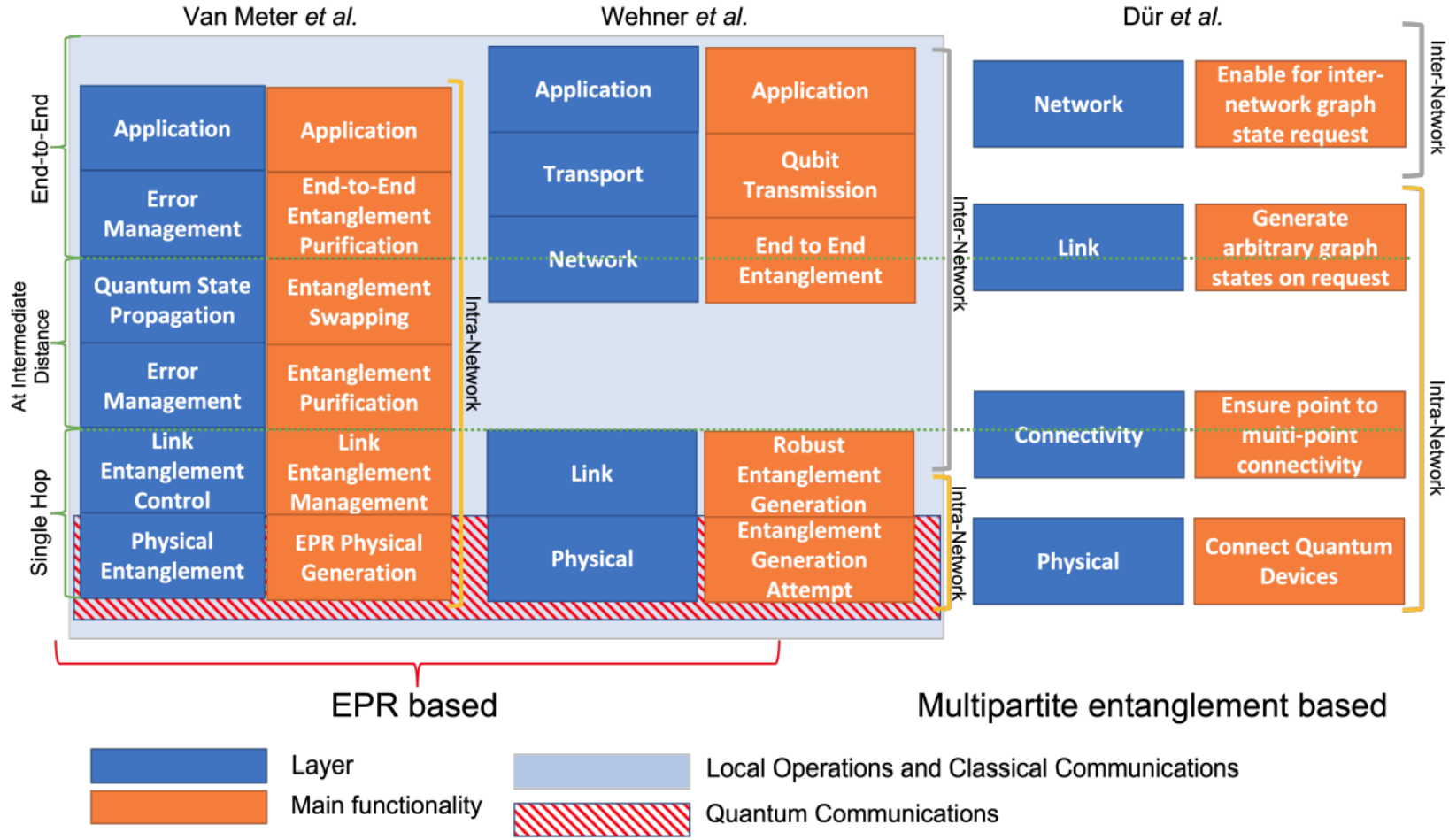
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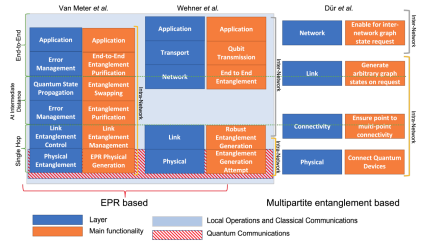
1. Neither stack is topped with applications, but either quantum algorithms or theory.
2. A layer in the stack could be considered an application for a particular community. For example, quantum error correction could be an application as could a quantum algorithm.
3. Both stacks are founded on physical aspects, which in some communities could also be the application.
4. Without judging either stack deficient, similar representations of other stacks include an application layer which is more user-oriented. As an example, refer to the Open Systems Interconnect (OSI) network model

Quantum Networking Stacks



Comparison of three quantum network stacks [4]

Quantum Networking Stacks



Comparison of three quantum network stacks [4]

1. The arXiv paper arXiv:2506.12195v1 from 2025-06-13 called *OSI Stack Redesign for Quantum Networks: Requirements, Technologies, Challenges, and Future Directions* describes a 9-layer, quantum-converged OSI stack.

Applications of Quantum Information

Sensing

- spatial measurements (e.g. translation, rotation)
- metrology of physical quantities (e.g. electric field, gravity)
- imaging (e.g. ghost imaging, computational imaging)

Computing

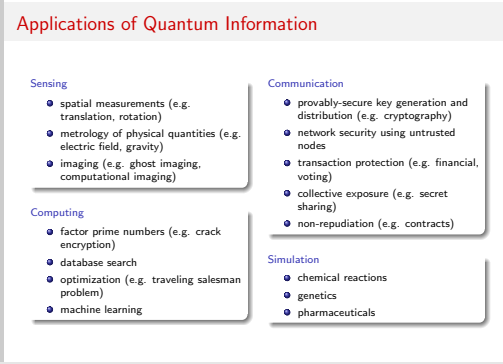
- factor prime numbers (e.g. crack encryption)
- database search
- optimization (e.g. traveling salesman problem)
- machine learning

Communication

- provably-secure key generation and distribution (e.g. cryptography)
- network security using untrusted nodes
- transaction protection (e.g. financial, voting)
- collective exposure (e.g. secret sharing)
- non-repudiation (e.g. contracts)

Simulation

- chemical reactions
- genetics
- pharmaceuticals



1. The title indicates that these can be considered applications, which might be valid in some contexts.
2. These could be also considered general capabilities that could be used in the service of applications or objectives.
3. The question of what constitutes an application largely depends on the audience.

Enablers to Objectives

Quid nostra refert? (Of what importance is it to us?)

Enablers

- single photon light source
- photon number resolving detector
- low-loss classical component
- photonic integrated circuit

Applications

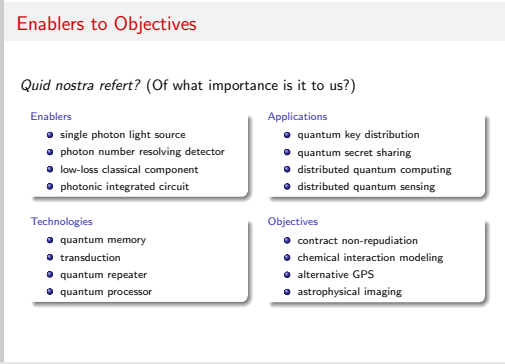
- quantum key distribution
- quantum secret sharing
- distributed quantum computing
- distributed quantum sensing

Technologies

- quantum memory
- transduction
- quantum repeater
- quantum processor

Objectives

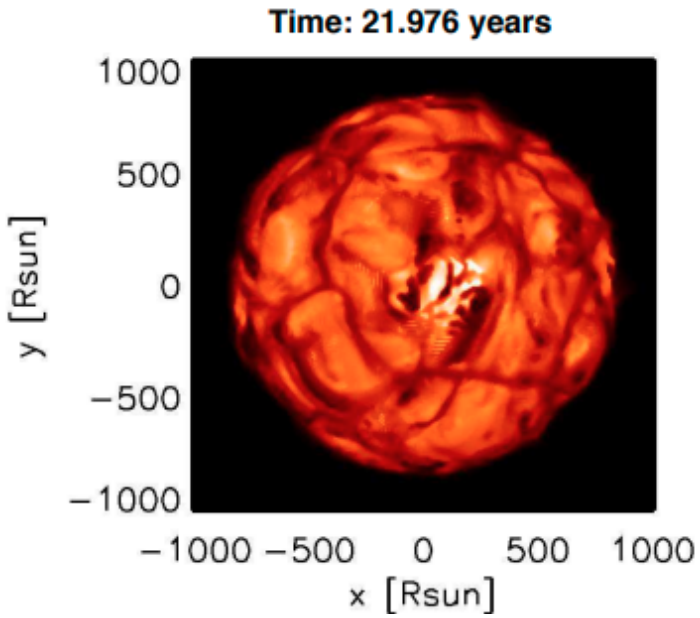
- contract non-repudiation
- chemical interaction modeling
- alternative GPS
- astrophysical imaging



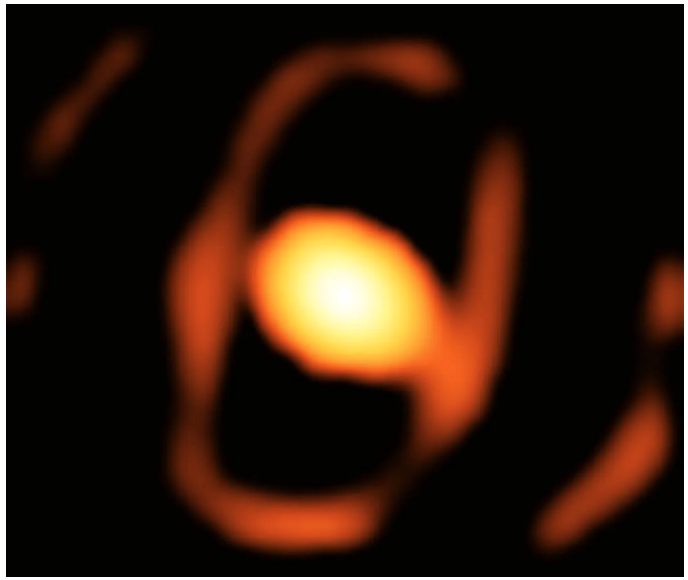
1. Categorization as enabler, technology, application or objective is a loose distinction for capabilities and levels of exposure.
2. Consider an analogy. A transistor and light-emitting diode are **enablers**. Many transistors and other electronic elements comprise a graphical processing unit (GPU) which could be considered a **technology**. A system of a GPU with other electronics and a display could yield the **application** of a workstation computer. (An unused computer could also be called an overly-engineered paperweight.) An **objective** is a goal to be accomplished, which could be done in different ways using different means (with one being a computer), which has its own costs and benefits for a particular objective,
3. Even the items identified as Objectives could be further refined into specific goals and use cases.

Astrophysical Imaging

Ex astris, scientia (From the stars, knowledge)—Starfleet Academy motto



Simulation of red super giant [7]

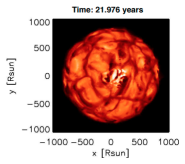


"First-ever close-up of a star [WOH G64] in another galaxy" [8]

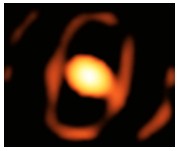
The simulation was used in the 2010 Interferometric Imaging Beauty Contest [9] to represent Gamma Crucis (γ Cru) at ~ 89 light years away. The WOH G64 close-up is from actual *interferometric imaging* of the star at 1.6×10^5 light years away.

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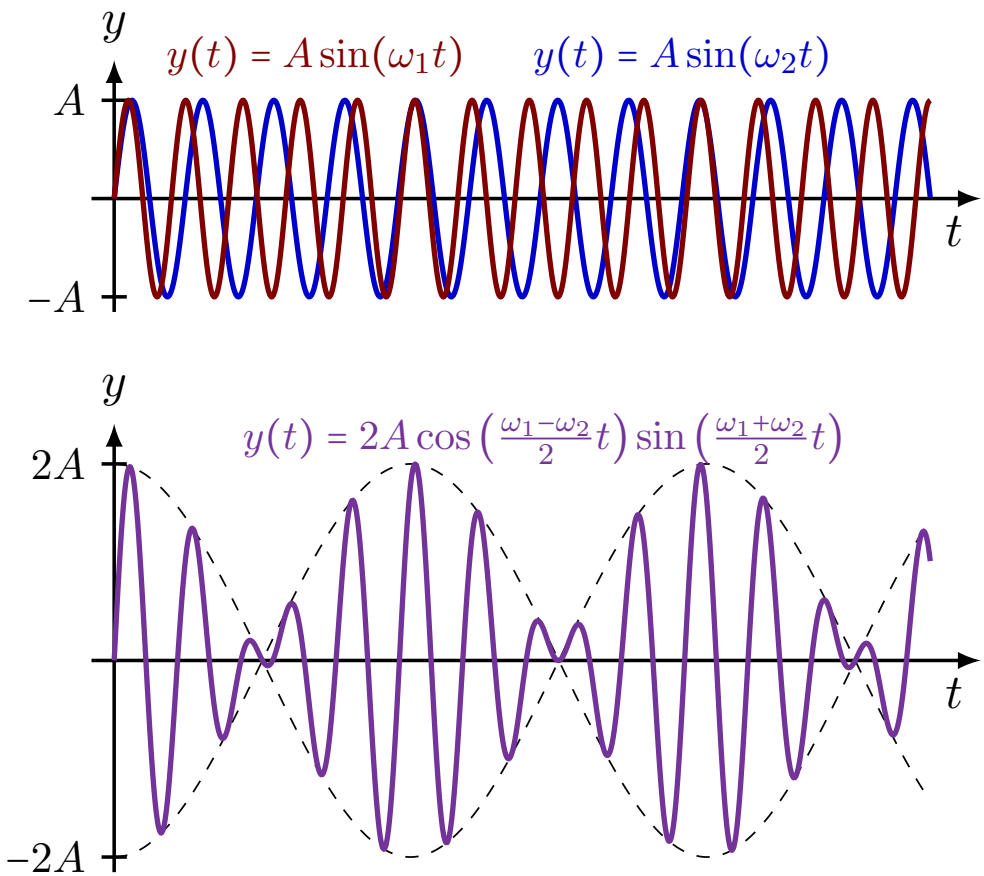
1. Imaging the structure of a star, other than Earth's sun, is at the edge of or beyond the capability of current optical sensing systems.
2. The Milky Way Galaxy is about 8.7×10^4 lightyears in diameter, about 1×10^3 lightyears thick at the spiral arms, and about 1.6×10^4 lightyears thick at the center's bulge.
3. A fun introduction about our place in the Milky Way and it's place in the universe is via The Galaxy Song by Monty Python.
4. The intergalactic image of the 'behemoth star' in the Large Magellanic Cloud was taken at the European Southern Observatory's Very Large Telescope Interferometer (VLTI) capturing light from four telecscopes.

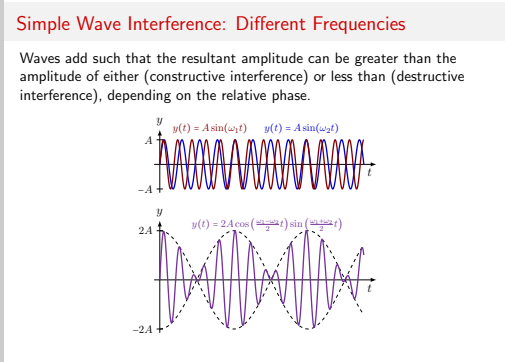
An electromagnetic wave equation

$$\frac{\partial}{\partial t}u(\mathbf{r},t) = c^2\nabla^2 u(\mathbf{r},t)$$

Simple Wave Interference: Different Frequencies

Waves add such that the resultant amplitude can be greater than the amplitude of either (constructive interference) or less than (destructive interference), depending on the relative phase.

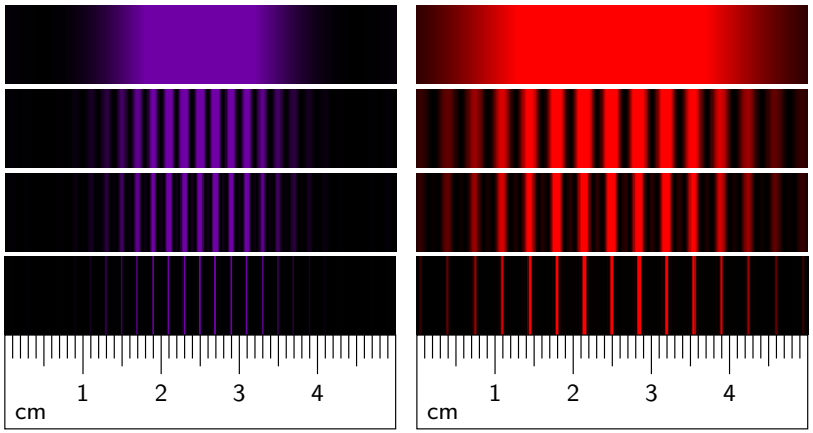




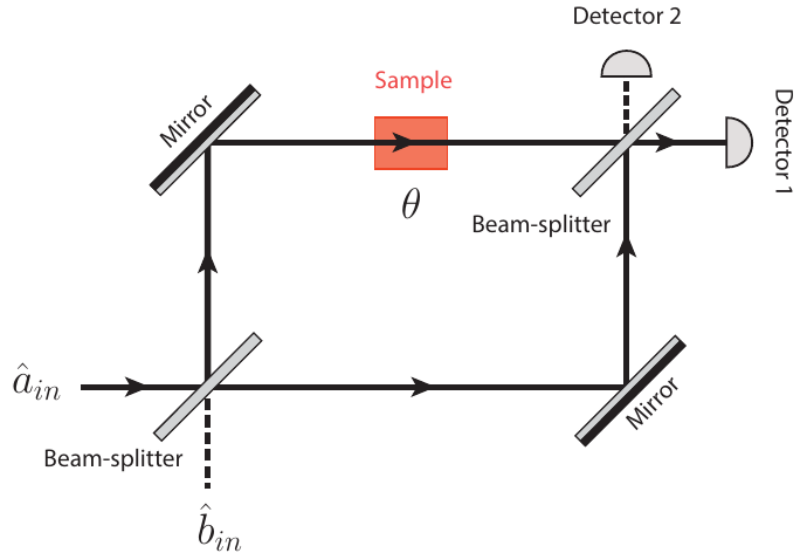
1. $y(t; \lambda, T, \phi) = y_{\text{max}} \sin(kx - \omega t - \phi)$ is the general form of a one-dimensional, sinusoidal wave with a maximum amplitude of y_{max} and wavelength $\lambda \equiv 2\pi/k$ traveling in space to the right and oscillating with a period $T \equiv 2\pi/\omega$, that began with a particular phase ϕ.
2. The example curves represent a single point in space ($x = 0$) for waves with different periods (which means different frequencies), both beginning with $\phi = 0$.
3. Two sinusoidal waves with the same wavelength, oscillatory period, and amplitude but non-zero phase difference $\Delta\phi = \phi_2 - \phi_1 \neq 0$ would obtain another sinusoidal wave whose maximum resultant amplitude is less than twice the amplitude of either.
4. Complete destructive interference—meaning a resultant amplitude of zero—occurs for $\Delta\phi = \pi$.

Interference from Slits and Splitters

"[I]nterference is always associated with the intrinsic indistinguishability of [. . .] paths" [13]

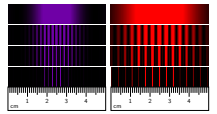


Interference patterns from 1, 2, 3, and 9 slits (from top to bottom) for $\lambda = 400\,\text{nm}$ (left) and $\lambda = 700\,\text{nm}$ (right) input to slits of width $10\,\mu\text{m}$ separated by $10\,\text{mm}$ at a distance of $0.5\,\text{m}$ from the detection plane

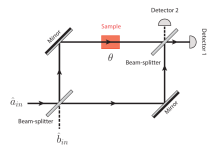


Two-port Mach-Zehnder interferometer [12]

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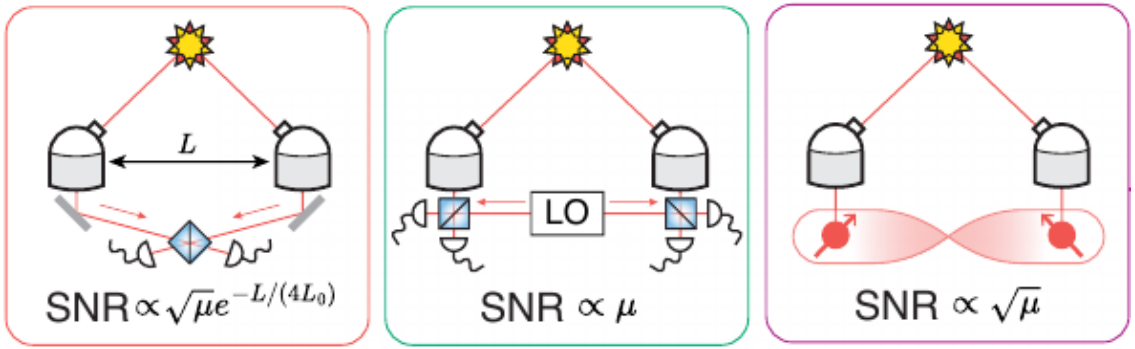
1. Observe from the slit patterns that (1) larger wavelengths spread (diffract) more than shorter wavelengths from the same slit width, (2) a bright region falls off from its center, in a similar fashion as the entire pattern falls off from the central bright region, and (3) more slits sharpen the bright locations and widen the dark regions between.
2. Detections from the Mach-Zehnder interferometer (MZI) would have complementary patterns due to conservation of energy.
3. Consider a single photon injected at one input port (e.g., $\hat{a}_{\text{in}}$) of the MZI and another *indistiguishable* photon is injected at the other port (e.g., $\hat{b}_{\text{in}}$), and the paths are identical (e.g., $\theta = 0$). The Hong-Ou-Mandel effect is the observation that without losses, there will never be a single photon detected at each detector, but only two photons in one detector with zero in the other, or vice versa.

An Introduction of Interferometric Imaging

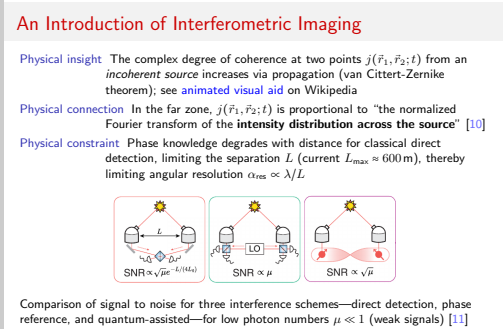
Physical insight The complex degree of coherence at two points $j(\vec{r}_1, \vec{r}_2; t)$ from an *incoherent source* increases via propagation (van Cittert-Zernike theorem); see animated visual aid on Wikipedia

Physical connection In the far zone, $j(\vec{r}_1, \vec{r}_2; t)$ is proportional to "the normalized Fourier transform of the **intensity distribution across the source**" [10]

Physical constraint Phase knowledge degrades with distance for classical direct detection, limiting the separation L (current $L_{\text{max}} \approx 600\,\text{m}$), thereby limiting angular resolution $\alpha_{\text{res}} \propto \lambda / L$



Comparison of signal to noise for three interference schemes—direct detection, phase reference, and quantum-assisted—for low photon numbers $\mu \ll 1$ (weak signals) [11]



1. The "essence of coherence" is that "the interfering beams possess some statistical similarity"—Emil Wolf in *Introduction to the Theory of Coherence and Polarization of Light* [10].
2. The van Cittert-Zernike theorem provides a path for indirect measurement of the intensity of a far away object that is unresolved by conventional means.
3. Factors related to separation distance that degrade phase knowledge are fluctuations in the medium (e.g., atmospheric turbulence for free space propagation, temperature and mechanical variations that change the properties of fiber waveguides), interactions with particles and impurities, signal loss reducing the signal-to-noise ratio, etc.
4. When there are enough signal photons, namely $\mu_{\mathrm{sig}} \ggg 1$, standard methods may be sufficient for detection and characterization.

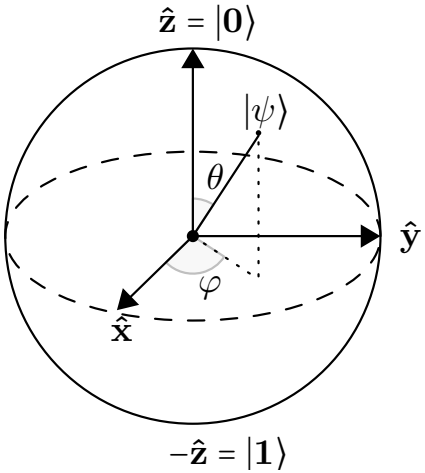
A quantum wave equation

$$i\hbar\frac{\partial}{\partial t}\Psi(\mathbf{r},t) = \left[-\frac{\hbar^2}{2m}\nabla^2 + V(\mathbf{r})\right]\Psi(\mathbf{r},t)$$

A Superposition of What?

In the abstract, a qu**b**it is a complex, linear superposition of two (exclusive and complete) states of a system in a two-dimensional Hilbert space — **not** *a vector in three dimensions*

$$|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle = \alpha\begin{pmatrix}1\\0\end{pmatrix} + \beta\begin{pmatrix}0\\1\end{pmatrix}$$



Atomic (spin)

$$|\psi\rangle_{\text{electron-spin}} = \alpha\,|\uparrow\rangle_{\text{el}} + \beta\,|\downarrow\rangle_{\text{el}}$$
$$|\psi\rangle_{\text{nuclear-spin}} = \alpha\,|\uparrow\rangle_{\text{nuc}} + \beta\,|\downarrow\rangle_{\text{nuc}}$$

Photonic (polarization and number)

$$|\psi\rangle_{\text{photon-pol}} = \alpha\,|\nwarrow\rangle_{\text{ph}} + \beta\,|\nearrow\rangle_{\text{ph}}$$
$$|\psi\rangle_{\text{photon-num}} = \alpha\,|0\rangle_{\text{ph}} + \beta\,|1\rangle_{\text{ph}}$$

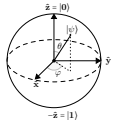
Electronic (charge, current)

$$|\psi\rangle_{\text{chg}} = \alpha\,|0\rangle_{\text{chg}} + \beta\,|2e\rangle_{\text{chg}}$$
$$|\psi\rangle_{\text{curr}} = \alpha\,|\circlearrowleft\rangle_{\text{curr}} + \beta\,|\circlearrowright\rangle_{\text{curr}}$$

A Superposition of What?

In the abstract, a qubit is a complex, linear superposition of two (exclusive and complete) states of a system in a two-dimensional Hilbert space — **not** *a vector in three dimensions*

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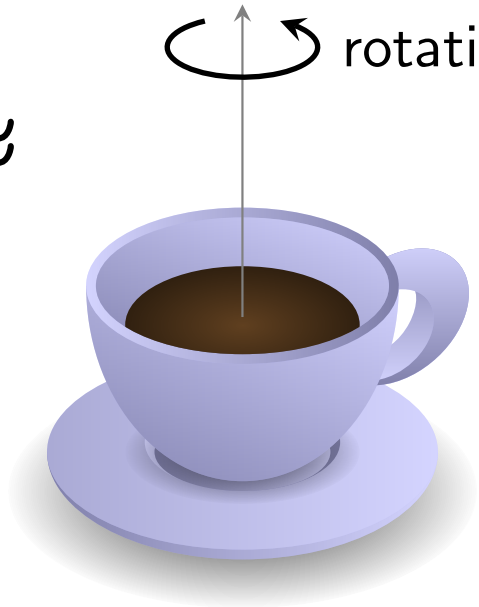
$$|\psi\rangle_{\text{chg}} = \alpha|0\rangle_{\text{chg}} + \beta|2e\rangle_{\text{chg}}$$
$$|\psi\rangle_{\text{curr}} = \alpha|\circlearrowright\rangle_{\text{curr}} + \beta|\circlearrowleft\rangle_{\text{curr}}$$

1. A qubit is a simple superposition that can bring out several ideas for quantum states.
2. The coefficients of basis states are complex probability *amplitudes* that are combined coherently, keeping phase information.
3. A single quantum state is not an observable.
4. A quantum state can be encoded in basis states from different physical systems and from different degrees of freedom in the same system.
5. A quantum state can be transferred from one physical system to another, for example $|\psi\rangle_{\text{photon-num}} \to |\psi\rangle_{\text{electron-spin}} \to |\psi\rangle_{\text{nuclear-spin}}$.

Planck's Call to Action

The difference between classical physics and quantum physics is Planck's constant $\hbar$ (technically, $i\hbar$).

$\hbar \approx$



rotating a cup a billionth of a turn . . .

. . . over the lifetime of the universe!

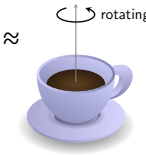


$\hbar \equiv 1.054\,571\,817 \times 10^{-34}\ \mathrm{J\,s}$

Planck's Call to Action

The difference between classical physics and quantum physics is Planck's constant $\hbar$ (technically, $i\hbar$).

$\hbar \approx$



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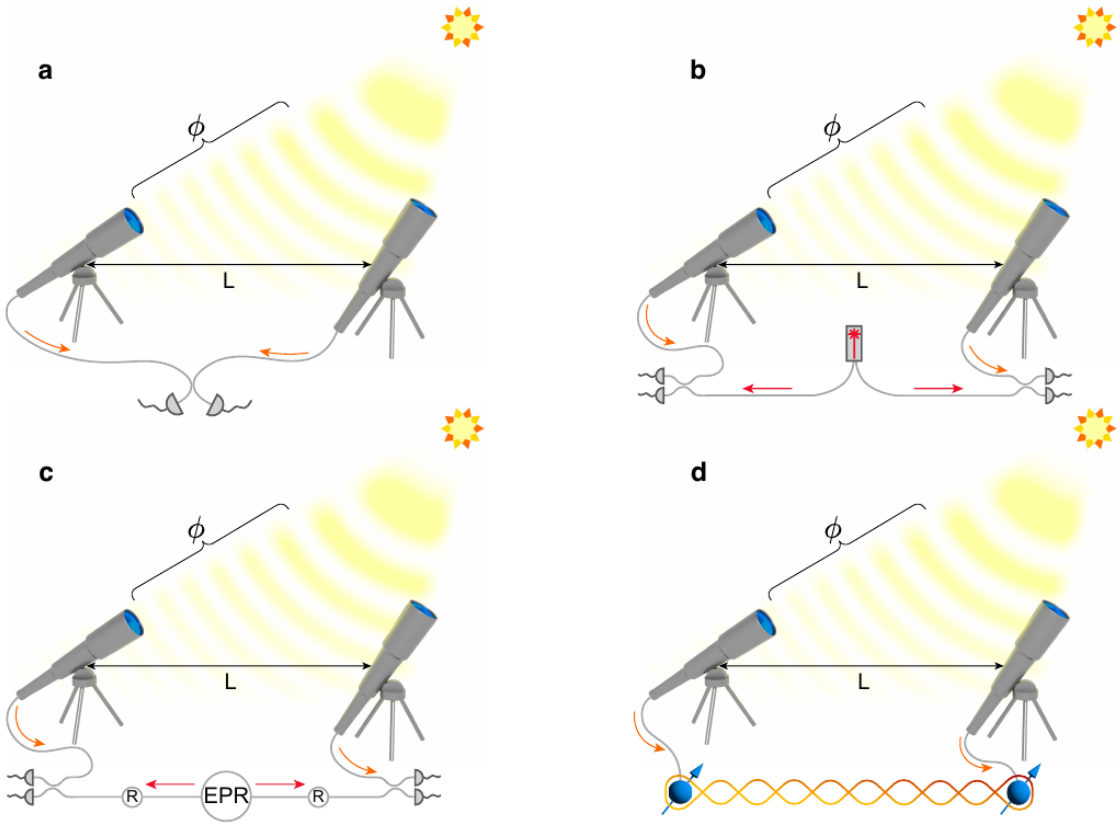
1. Quantum effects are not readily noticeable in our common experiences due to the smallness of Planck's constant.
2. Entertaining and informative science fantasies by George Gamow are collected in *Mr. Tompkins in Paperback*. His literary device modifies physical constants so that subtle effects are more obvious. In the present case, Planck's constant is made larger in the stories *Quantum Billiards*, *Qauntum Jungles*, and *The Gay Tribe of Electrons*.

The Delicate, Competitive Dance of Quantum Information

- Isolation
  - Generated quantum states are fragile
  - Noise destroys, or at least complicates, their coherent nature
  - Segregation from environment and other systems is paramount . . .
- Interaction
  - . . . except splendid isolation leaves no means for control
  - Manipulation and detection of quantum states is necessary for utility
  - Quantum systems need to influence one another and another and . . .
- Interconnection
  - . . . a completely other
  - Collections of heterogeneous quantum systems must inter-communicate quantum information

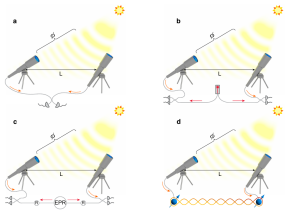
Quantum Information for Very Long Baseline Interferometry (VLBI)

Four VLBI Schemes



Top row schemes are purely classical—(a) direct detection, (b) local oscillator-assisted– and the bottom are quantum-enhanced—(c) Gottesman-Jennewein-Croke (GJC), (d) Khabiboulline–Borregaard–de Greve–Lukin (KBGL) [11]

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1. The GJC protocol is considered the first non-local quantum scheme for VLBI.
2. VLBI measures the complex degree of coherence from an object, which is related to the phase difference ϕ due to the path length difference to the collectors.

Recent Demonstration of KBGL Protocol

Entanglement Assisted Non-local Optical Interferometry in a Quantum Network

P.-J. Stas,[1,*] Y.-C. Wei,[1,*] M. Sirotin,[1,2,*] Y. Q. Huan,[1] U. Yazlar,[1,3]
F. Abdo Arias,[1] E. Knyazev,[1] G. Baranes,[1,2] B. Machielse,[4,5] S. Grandi,[6] D. Riedel,[5]
J. Borregaard,[1] H. Park,[1,7] M. Lončar,[4] A. Suleymanzade,[1,8] and M. D. Lukin[1,†]

[1] *Department of Physics, Harvard University, Cambridge, Massachusetts 02138, USA*
[2] *Department of Physics and Research Laboratory of Electronics, Massachusetts Institute of Technology, Cambridge, Massachusetts 02138, USA*
[3] *Division of Materials Science & Engineering, Boston University, Boston, Massachusetts 02215, USA*
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[6] *ICFO-Institut de Ciencies Fotoniques, The Barcelona Institute of Science and Technology, 08860 Castelldefels (Barcelona), Spain*
[7] *Department of Chemistry and Chemical Biology, Harvard University, Cambridge, Massachusetts 02138, USA*
[8] *Department of Physics, University of California, Berkeley, California 7300, USA*

(Dated: September 16, 2025)

arXiv:2509.09464

Enablers to Objectives

Quid nostra refert? (Of what importance is it to us?)

Enablers

- single photon light source
- **photon number resolving detector**
- **low-loss classical component**
- photonic integrated circuit

Applications

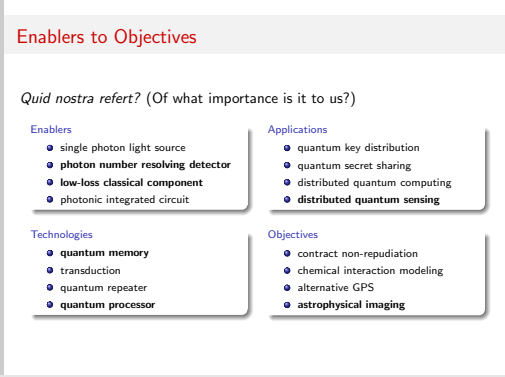
- quantum key distribution
- quantum secret sharing
- distributed quantum computing
- **distributed quantum sensing**

Technologies

- **quantum memory**
- transduction
- quantum repeater
- **quantum processor**

Objectives

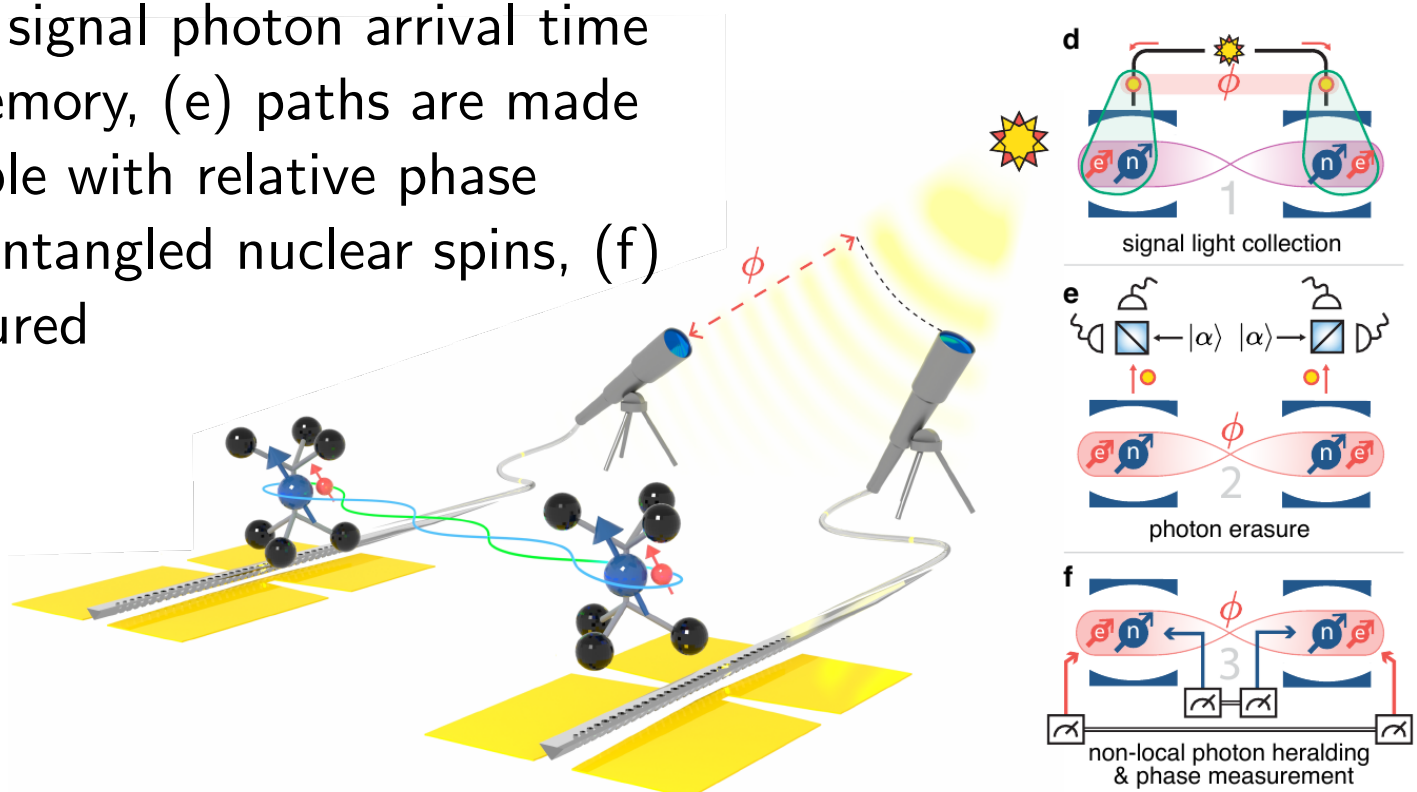
- contract non-repudiation
- chemical interaction modeling
- alternative GPS
- **astrophysical imaging**



1. The highlighted items are important for the KBGL protocol to meet an objective of astrophysical imaging of weak signals.

KBGL VLBI Protocol Overview

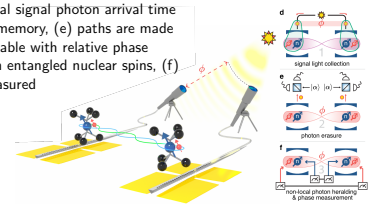
Matter memories at each collection site (Left and Right) are pre-entangled, (d) weak, thermal signal photon arrival time is stored in memory, (e) paths are made indistinguishable with relative phase imprinted on entangled nuclear spins, (f) phase is measured



KBGL VLBI scheme with pre-entangled memories [11]

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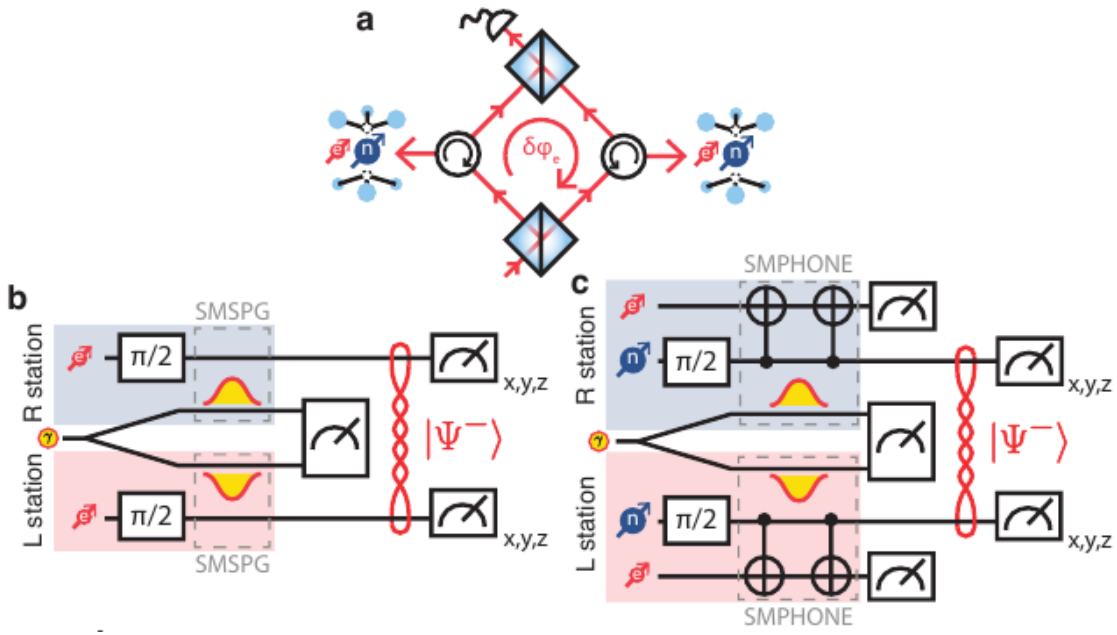


KBGL VLBI scheme with pre-entangled memories [11]

1. Entangled memories are a pre-condition of the protocol. While the KBGL protocol uses a particular method to pre-entangle the memories, other methods such as a fountain of entangled photons (also known as entanglement distribution) could be used with techniques to transfer the photons' entangled state to memories.

Entangle Matter Memories

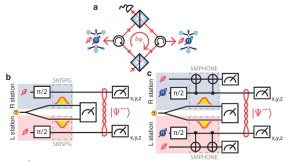
A reference pulse γ is shared between the matter memories of two stations via (a) Mach-Zehnder interferometer that (b) entangles the electron spins using single-mode spin-photon (SMSPG) gates, and subsequently (c) entangles the nuclear spins using single-mode photon-nucleus entanglement (SMPHONE) gates.



Entangle nuclear memories via electron spin entanglement [11]

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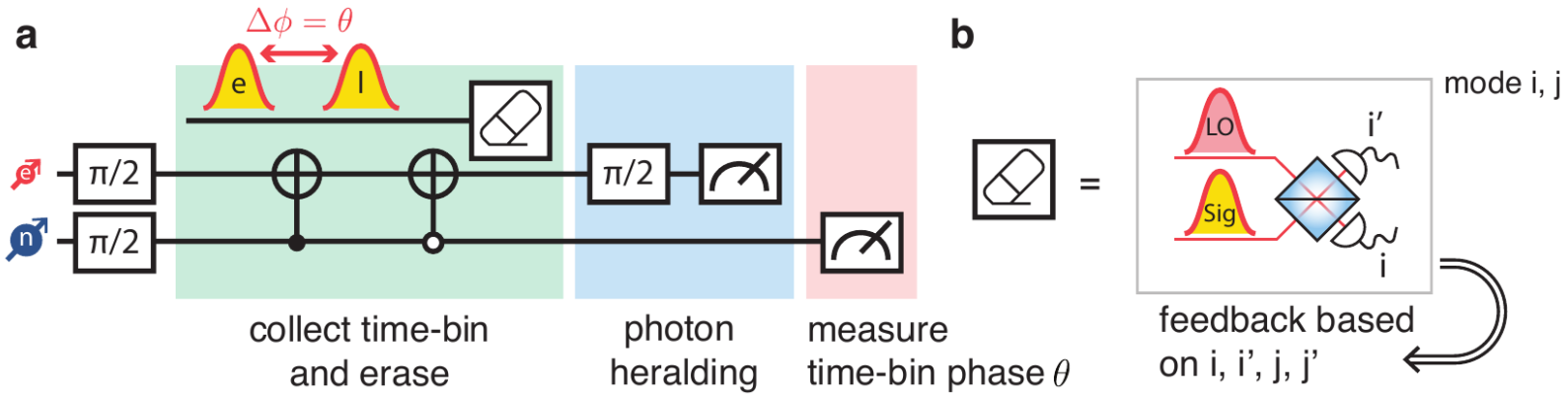


Entangle nuclear memories via electron spin entanglement [11]

1. This part of the protocol begins with light from a system source, not photons from the object.
2. The method first encodes entanglement in the electron spins of silicon vacancy center matter memories, then transfers the entangled quantum state to the nuclear spins via rotation, controlled not gates, and measurement.

Remove Which-Path Information: Erasure

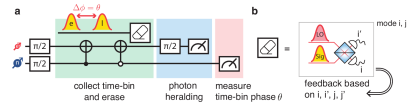
Weak signal state emphasizing spatial separation:
$|\psi_{\text{sig}}\rangle \sim |0_L 0_R\rangle + \sqrt{\mu_{\text{sig}}/2}\left(|0_L 1_R\rangle + e^{i\phi}|1_L 0_R\rangle\right)$
Spatial separation yields temporal separation with a phase difference $\Delta\phi$.
Time bins are stored in the nuclear spin of the entangled memory using a binary encoding scheme with path information erased.
System state after erasure:
$|\psi_{\text{sys}}\rangle = |\downarrow\rangle_{\text{el}}|+\rangle_{\text{nuc}} + \sqrt{\mu_{\text{sig}}/2}\left[|+\rangle_{\text{el}}|+\rangle_{\text{nuc}} + e^{i\theta}\left(|+\rangle_{\text{el}}|\downarrow\rangle_{\text{nuc}} + |-\rangle_{\text{el}}|\uparrow\rangle_{\text{nuc}}\right)/\sqrt{2}\right]$
Apply Pauli Z operator $\hat{\sigma}_z$ on the nuclear spins to obtain relative phase θ.



Teleportation circuit (green part of a with b) with erasure operation [11]

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Teleportation circuit (green part of a with b) with erasure operation [11]

1. The weak signal state uses L and R to accentuate the spatial separation of the physical collections. The e and l notation emphasizes the temporal separation of the signals.
2. Optical path differences to the collectors yield early (e) and late (l) signals which produce the relative phase difference $(\Delta\phi = \phi_L - \phi_R \equiv \theta)$ to be measured, but they are then **distinguishable**.
3. Note that if one collector is taken as reference with respect to phase (e.g., $\phi_R = 0$), then $\Delta\phi = \phi \equiv \theta$.
4. The binary encoding scheme reduces the number of quantum memory resource required to store a time bin.
5. The arrival time-bin is encoded in the nuclear spin depending on the combined click pattern of the erasure circuits on both stations which removes the which-path information.

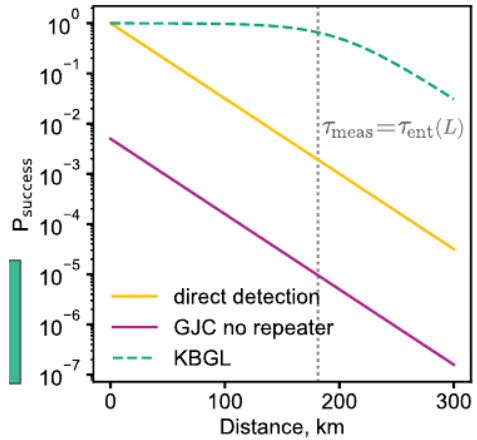
Measure of Quantum Advantage: Fisher Information

The Fisher information $\mathcal{F}_{\mathsf{I}}$ is identified as a signal-to-noise ratio $\varrho_{\mathsf{s:n}}$ metric, namely $\varrho_{\mathsf{s:n}} \equiv \sqrt{\mathcal{F}_{\mathsf{I}}}$.

A system will have its Fisher information reduced due to non-idealities, captured here in constants η_{erase}, η_{herald}, and $\bar{V}$. For the KBGL protocol, the Fisher information including imperfections is given by

$$\mathcal{F}_{\mathsf{I}} = \eta_{\mathsf{erase}}\eta_{\mathsf{herald}}\bar{V}^2\frac{\mu_{\mathsf{sig}}^2 e^{-2\mu_{\mathsf{sig}}}}{1-e^{-\mu_{\mathsf{sig}}}} \rightarrow \mathcal{F}_{\mathsf{I}} \propto \mu_{\mathsf{sig}} \text{ for } \mu_{\mathsf{sig}} \ll 1$$

which is the same scaling as classical direct detection.

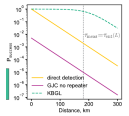


The KBGL protocol maintains significant distance advantage over direct detection [11]

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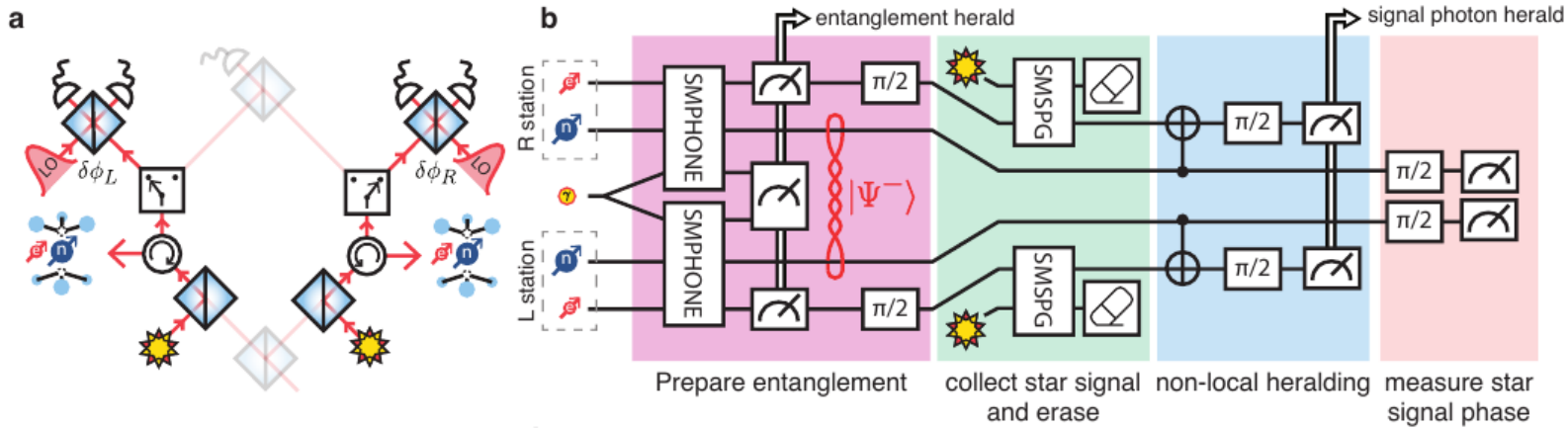
The KBGL protocol maintains significant distance advantage over direct detection [11]

1. A system with non-idealities can be characterized by constant parameters: (1) erasure and heralding efficiencies, η_{erase} and η_{herald} respectively, (2) gate errors, imperfect erasure, and reduced state fidelities captured together in a parameter $\bar{V}$.

KBGL VLBI Protocol Rearview

Key elements of the KBGL protocol:

- pre-entangled matter memories in cavities using MZI with local photons
- signal light time-bin arrival storage with path information erased for indistinguishability
- non-local heralding (not discussed) of signal photon to prepare for measurement
- phase measurement

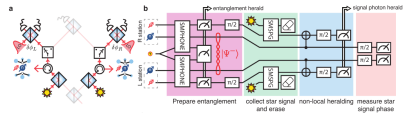


Compact summary of KBGL protocol [11]

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Compact summary of KBGL protocol [11]

1. Familiarity with quantum computing circuit diagrams and operations can help the reader interpret the algorithm presented in the paper.
2. Familiarity with physics principles (e.g., interference) and concepts (e.g., quantum states in varied physical embodiments) will also help the reader understand the preconditions and objectives.

A Holy Curiosity

> *The important thing is not to stop questioning. Curiosity has its own reason for existence. One cannot help but be in awe when he contemplates the mysteries of eternity, of life, of the marvelous structure of reality.* **It is enough if one tries merely to comprehend only a little of this mystery every day.** *Never lose a holy curiosity.*

— Albert Einstein, *Death of a Genius* in LIFE magazine (April 1955)

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> — Albert Einstein, *Death of a Genius* in LIFE magazine (April 1955)

1. Einstein was a genius of science whose sense of wonder and awe likely enspirited his quest for understanding. Experiencing mysteries did not seem to dull his capacity for science.

Mysteries of Eternity, of Life, of [. . .] Reality



Milky Way over Lake Powell (Tim Burt, Nikon D7000, AF-S Nikkor 20mm, F1.8, 15s)

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